POSTS AND TELECOMMUNICATIONS
INSTITUE OF TECHNOLOGY
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INTERNSHIP

REPORT

WoxionChat

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1 Overview

This week focused on the transition from architectural design to full-scale implementation. The primary achievement was the successful completion of the **AI API codebase**. We have bridged the gap between the fine-tuned LLM and the Django backend, enabling seamless, secure, and high-performance communication for both standard chat and autonomous agentic workflows.

2 Technical Accomplishments

2.1 AI API Implementation

The core API layer for AI interactions is now fully coded and verified:

- **RESTful Endpoints:** Developed comprehensive endpoints for stateless AI queries, allowing external modules to interact with the fine-tuned model.
- **WebSocket Integration:** Finalized the real-time communication layer using Django Channels, enabling streaming responses which significantly improve the perceived user experience.
- **Response Parsing:** Implemented robust output parsers to handle structured data from the model, ensuring the backend can reliably execute tool-calling commands.

2.2 Endpoint Security & Optimization

- **Authentication Guard:** Integrated JWT and API Key validation for all AI-related requests, ensuring only authorized sessions (verified Staff and Boss) can trigger expensive inference tasks.
- **Rate Limiting:** Implemented throttling mechanisms to prevent API abuse and manage GPU resource allocation effectively.
- **Caching Layer:** Deployed a Redis-based cache for frequent queries to reduce latency and redundant model computation.

2.3 Agentic Logic Execution

The previously designed Agentic system has been successfully translated into executable code:

- **Tool Execution Engine:** Built the handler that allows the AI to trigger real-world actions, such as fetching user session logs or querying the MongoDB document store.
- **State Management:** Developed a session-aware memory controller that tracks the "Thought Process" of the agent across multiple interaction turns.

3 Technology Stack Update

- **Frameworks:** Django REST Framework, Django Channels (WebSockets).
- **Execution:** Celery Workers for heavy Agentic reasoning tasks.
- **Communication:** JSON-RPC and REST protocols.

4 Progress Evaluation

- **Status:** On Track.
- **Observations:** Completing the API code marks the end of the core backend development phase. The system is now ready for frontend integration and comprehensive end-to-end testing.

5 Upcoming Milestone (Next Week)

1. **Frontend Chat Interface:** Integrating the newly finished AI API with the React/Vue frontend components.
2. **UAT (User Acceptance Testing):** Conducting initial testing sessions with "The Boss" role to verify the utility of administrative AI tools.
3. **Production Staging:** Setting up the staging environment for live model serving and API performance monitoring.